# PHYSICS

1) Gravel is dropped onto a conveyer belt at a rate of 0.5 kg/s. The extra force required in newton to keep the belt moving at 2 m/s is :-

(1) 1 N
(2) 2 N
(3) 4 N
(4) 0.5 N

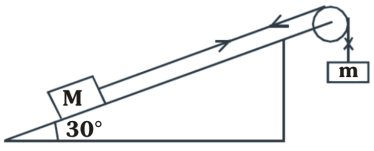
2) When a spring is subjected to 4N force its length is a metre and if 5N is applied, its length is b metre. If 9N is applied, its length is :-

(1) 4b – 3a
(2) 5b – a
(3) 5b – 4a
(4) 5b – 2a

3) A 1 kg object strikes a wall with velocity 1 m $s^{-1}$ at an angle of 60° with the wall and reflects at the same angle. If it remains in contact with wall for 0.1 s, then the force exerted on the wall is :-

(1) $30\sqrt{3}$ N
(2) Zero
(3) $10\sqrt{3}$ N
(4) $20\sqrt{3}$ N

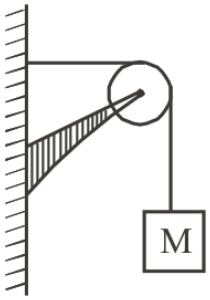
4) In the fig. mass M = 10 g is placed on frictionless plane. In order to keep it at rest, the value of mass m will be :




(1) 5 g
(2) 10 $\sqrt{3}$ g
(3) 0.10 g
(4) $\sqrt{3}$ g

5) A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as

shown in the figure. The force on the pulley by the clamp is given.



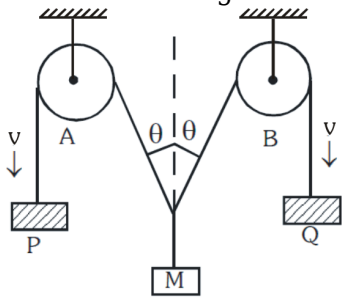

(1) $\sqrt{2}Mg$

(2) $\sqrt{2}mg$

(3) $\sqrt{(M+m)^2+m^2}\,g$

(4) $\sqrt{(M+m)^2+M^2}\,g$

6) In the fig., the ends P and Q of an unstrechable string move downward with uniform speed v.




Mass M moves upwards with speed.

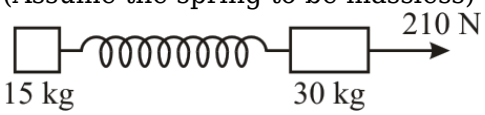
(1) v cosθ

(2) v/cosθ

(3) 2v cosθ

(4) 2/v cosθ

7) In the system shown, the mass 30 kg is pulled by a force 210 N. At the instant when the 15 kg mass has acceleration 6 m/s$^2$, the acceleration of 30 kg mass will be :-
(Assume the spring to be massless)




(1) 2 m/s$^2$

(2) 3 m/s$^2$

(3) 3.4 m/s$^2$

(4) 4 m/s$^2$

8) **Assertion :** The apparent weight of a person standing in a lift which moves upwards with uniform acceleration is always higher than his true weight.
**Reason :** The weight (on the earth surface) always acts downwards.

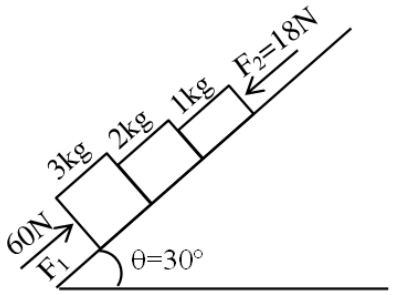
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

9) In the diagram shown, the normal reaction force between 2 kg and 1 kg is (Consider the surface, to be smooth): Given g = 10 $ms^{-2}$



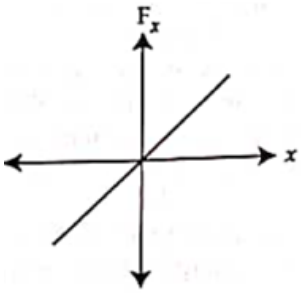

(1) 25 N

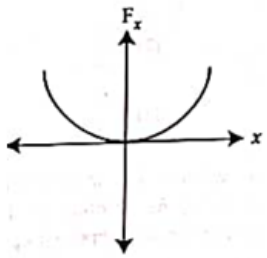
(2) 39 N

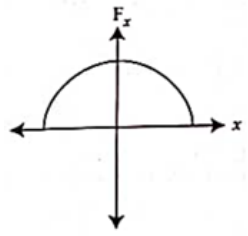
(3) 6 N

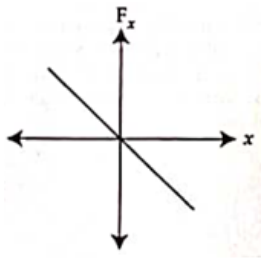
(4) 10 N

10) The restoring force of a spring with a block attached to the free end of the spring is represented by :

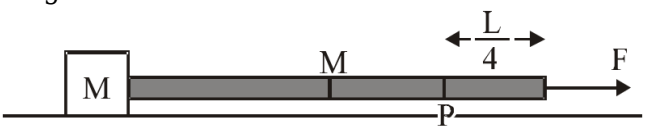
(1) 


(2) 


(3) 


(4) 

11) A block of mass M is pulled by a uniform chain of mass M tied to it by applying a force F at the other end of the chain. The tension force at a point located at a distance of one-fourth of the chain's length from the free end of chain will be:-




(1) $\frac{7F}{8}$

(2) $\frac{4F}{5}$

(3) $\frac{3F}{4}$

(4) $\frac{6F}{7}$

12) In Newton's second Law $\vec{F} = m\vec{a}$ (for constant mass m), $\vec{a}$ is the acceleration of the mass m with respect to

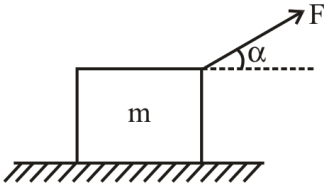
(1) any observer

(2) any inertial observer

(3) an observer at rest only

(4) an observer moving with uniform speed only

13) At the instant t = 0 a force F = kt (k is a constant) acts on a small body of mass m resting on a smooth horizontal surface. The time, when body leaves the surface is :-



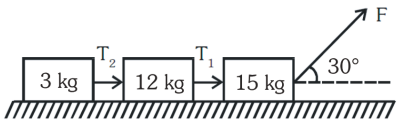

(1) mg k sin α

(2) k sin α/mg

(3) mg sin α/k

(4) mg/k sin α

14) The surface is frictionless, the ratio of $T_1$ and $T_2$ is :-




(1) $\sqrt{3}$ : 1

(2) 1 : $\sqrt{3}$

(3) 1 : 5

(4) 5 : 1

15) A tennis ball is dropped on the floor from a height of 20 m. It rebounds to a height of 5 m. The ball was in contact with the floor for 0.01 s. What was its average acceleration during the contact ?

($g = 10 m/s^2$)

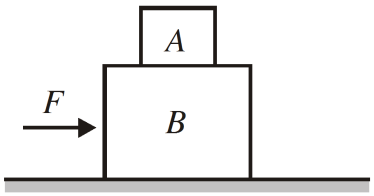
(1) 3000 $m/s^2$

(2) 2000 $m/s^2$

(3) 1000 $m/s^2$

(4) 500 $m/s^2$

16) A 10 N force is applied on B. All surfaces are smooth. Mass of A & B are 5 kg each. What is the acceleration of A?



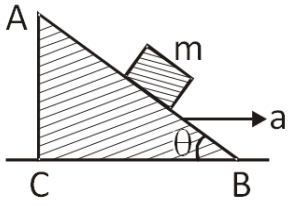

(1) 0 $m/s^2$

(2) 1 $m/s^2$

(3) 2 $m/s^2$

(4) 3 $m/s^2$

17) A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and θ for the block to remain stationary on the wedge is :-




(1) $a = \frac{g}{\text{cosec}\,\theta}$

(2) $a = \frac{g}{\sin\theta}$

(3) $a = g \cos\theta$

(4) $a = g \tan\theta$

18) A body moving with uniform velocity is stopped in 0.25 second by applying a retarding force of 200 Newton. The initial momentum of the body will be :-

(1) 50 N-s

(2) 100 N-s

(3) 150 N-s

(4) Zero

19) A boy standing on a weighing machine notices his weight as 400 N. When he suddenly jumps upward, the weight shown by the machine becomes 600 N. The acceleration with which the boy jumps up is :- ($g = 10 m/s^2$)

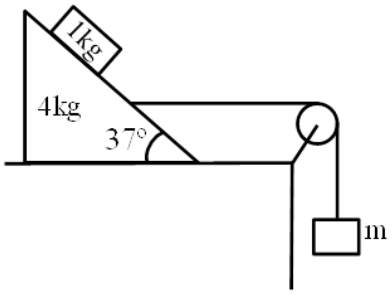
(1) 5 $ms^{-2}$

(2) 3.4 $ms^{-2}$

(3) 6 $ms^{-2}$

(4) 9.8 $ms^{-2}$

20) If all surfaces are frictionless then what will be the value of mass 'm' so that 1 kg block does not slip:




(1) 5 kg

(2) 10 kg

(3) 15 kg

(4) None

21) Two weights $w_1$ and $w_2$ are connected by a light thread which passes over a light smooth pulley. If the pulley is raised upwards with an acceleration equal to g, then the tension in the thread will be :-

(1) $\frac{2w_1w_2}{w_1+w_2}$

(2) $\frac{w_1w_2}{w_1+w_2}$

(3) $\frac{4w_1w_2}{w_1+w_2}$

(4) $\frac{4w_1w_2}{w_1-w_2}$

22) When a horse pulls a wagon, the force that causes the horse to move forward is the force :-

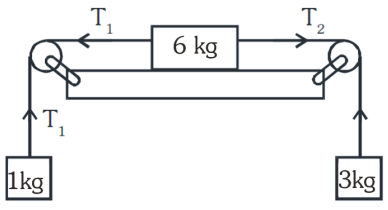
(1) He exerts on the wagon

(2) The wagon exerts on him

(3) The ground exerts on him

(4) He exerts on the ground

23) Three masses of 1 kg, 6 kg and 3 kg are connected to each other with threads and are placed on a table as shown in figure. What is the acceleration with which the system is moving ? (Take g = 10 $m/s^2$)



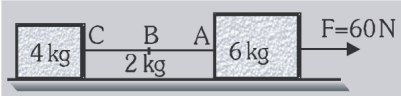
(1) Zero

(2) 2 $m/s^2$

(3) 4 $m/s^2$

(4) 3 $m/s^2$

24) Two blocks of masses 6 kg and 4 kg connected by a rope of mass 2 kg are resting on a frictionless floor as shown in the following figure. If a constant force of 60 N is applied to 6 kg block, then the tension in the rope at points A, B and C are respectively given by :-



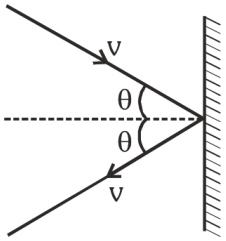

(1) 60 N, 60 N, 60 N

(2) 30 N, 25 N, 20 N

(3) 20 N, 25 N, 30 N

(4) 20 N, 20 N, 20 N

25) A water jet, whose cross sectional area is 'a' strikes a wall making an angle 'θ' with the normal and rebounds elastically. The velocity of water of density 'd' is v. Force exerted on wall is :-




(1) 2 $av^2d\cos\theta$

(2) 2 $av^2d\sin\theta$

(3) 2 $avd\cos\theta$

(4) $avd\cos\theta$

26) A balloon with mass 'm' is descending down with an acceleration 'a' (where a < g). How much mass should be removed from it so that it starts moving up with an acceleration 'a' ? (Assume that it's volume does not change)

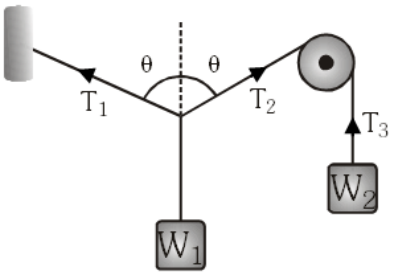
(1) $\frac{2ma}{g+a}$

(2) $\frac{2ma}{g-a}$

(3) $\frac{ma}{g+a}$

(4) $\frac{ma}{g-a}$

27) In the following figure, the pulley is massless and frictionless. The relation between $T_1$, $T_2$ and $T_3$




will be :-

(1) $T_1 = T_2 \neq T_3$

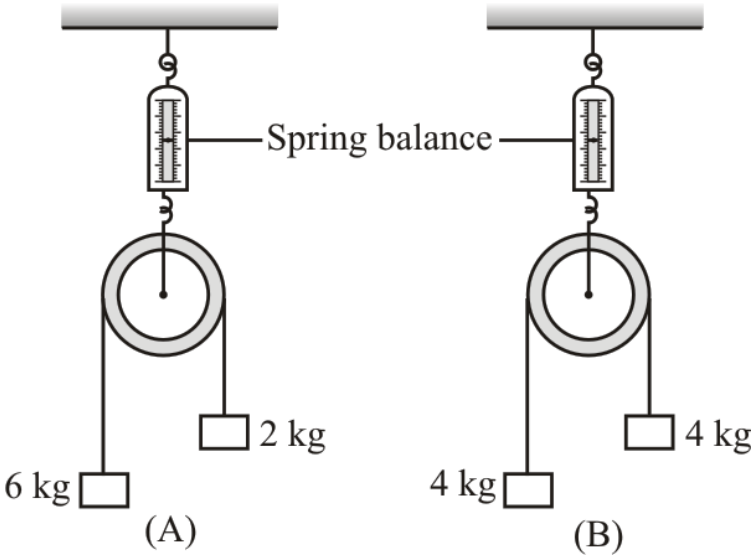
(2) $T_1 \neq T_2 = T_3$

(3) $T_1 \neq T_2 \neq T_3$

(4) $T_1 = T_2 = T_3$

28)

Referring figure A and B :-




(1) reading of spring balance in (A) is 8 g and in (B) is also 8 g

(2) reading of spring balance in (A) is 8 g and in (B), it is less than 8 g

(3) reading in (A) is less than 8 g and in (B), it is 8 g

(4) reading in both (A) and (B) is less than 8 g

29) There are two bodies A & B of same mass, body A is at rest while body B is under going uniform motion, which is correct statements?

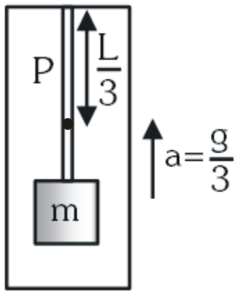
(1) Inertia of A > inertia of B.

(2) Inertia of B > inertia of A.

(3) Inertia of A = inertia of B.

(4) Either $1^{st}$, $2^{nd}$ or $3^{rd}$ is correct, depending upon the shape of body.

30) A block of mass m is attached to a uniform string of mass 3m and length L as shown in the figure. The lift is moving upward with acceleration a = g/3. the tension in the string at point P is




(1) 2 mg

(2) 4 mg

(3) 3 mg

(4) mg

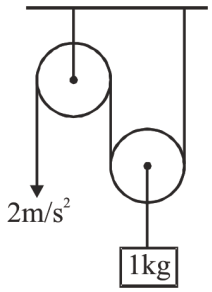
31) A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is :

(1) along northward

(2) along north-east

(3) along south-west

(4) along eastward




32) If string is pulled with acceleration of 2 $m/s^2$ then acceleration of block is.

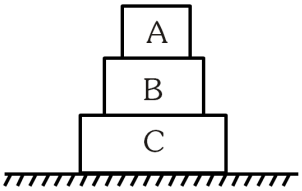
(1) 4 $m/s^2$

(2) 2 $m/s^2$

(3) 1 $m/s^2$

(4) 8 $m/s^2$

33) Three blocks are placed as shown in figure. Mass of A, B and C are $m_1$, $m_2$, and $m_3$ respectively. The force exerted by block 'C' on 'B' is :-



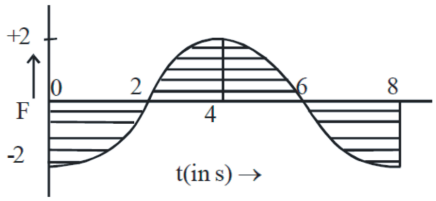

(1) $m_1g$

(2) $(m_1 + m_2)g$

(3) $m_2g$

(4) $(m_1 + m_2 + m_3)g$

34) Force-time graph for the motion of a body is shown in fig. Change in linear momentum between 0 to 8 s is :-




(1) Zero

(2) 4 N-s

(3) 8 N-s

(4) None of these

35) A cart is moving with a velocity 20 m/s. Sand is being dropped into the cart at the rate of 50 kg/min. The force required to move the cart with constant velocity will be :-

(1) 50 N

(2) 30.33 N

(3) 26.45 N

(4) 16.66 N

36) A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10\ m/s^2$) nearly :

(1) 0 kg m/s

(2) 4.2 kg m/s

(3) 2.1 kg m/s

(4) 1.4 kg m/s

37) A sphere is accelerated upwards with the help of a cord whose breaking strength is five times its weight. The maximum acceleration with which the sphere can move up without cord breaking is :-

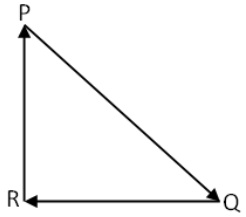
(1) 4g

(2) 3g

(3) 2g

(4) g

38) A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will :



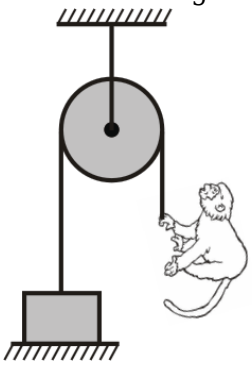

(1) increase

(2) decrease

(3) remain constant

(4) change according to the smallest force $\overrightarrow{QR}$

39) A monkey weighing 10 kg is climbing up a light rope and frictionless pulley attached to 15 kg mass at other end as in figure. In order to raise the 15 kg mass off the ground the monkey must



climb-up :-

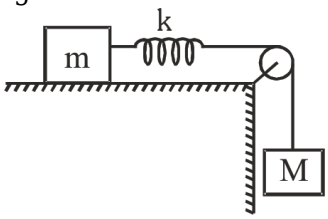
(1) with constant acceleration g/3.

(2) with an acceleration greater than g/2.

(3) with an acceleration greater than g/4.

(4) It is not possible because weight of monkey is lesser than the block.

40) Two masses, M and m are connected together by a pulley two strings and a stretched spring of force constant k as shown. Assume that string, pulley and spring all are massless and surface below




m is smooth. The amount by which the spring is stretched is :-

(1) $\frac{mMg}{k(m+M)}$

(2) $\frac{2mMg}{k(m+M)}$

(3) $\frac{Mg}{k}$

(4) $\frac{(m+M)g}{2k}$

41) The velocity acquired by a mass m in travelling a certain distance d starting from rest under the action of a constant force is directly proportional to :-

(1) $\sqrt{m}$

(2) m°

(3) $\frac{1}{\sqrt{m}}$

(4) m

42) A truck is stationary and has a bob suspended by a light string, in a frame attached to the truck. The truck, suddenly moves to the right with an acceleration of a. The pendulum will tilt :-

(1) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}\left(\frac{g}{a}\right)$

(2) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}\left(\frac{a}{g}\right)$

(3) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}\left(\frac{a}{g}\right)$

(4) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}\left(\frac{g}{a}\right)$

43) A material body A of mass $m_1$ exerts a force on another material body B of mass $m_2$. If the acceleration of B be $a_2$, the magnitude of the acceleration of A is :-

(1) Zero

(2) $m_2a_2/m_1$

(3) $m_1a_2/m_2$

(4) $a_2$

44)

The unit vector perpendicular to both vector $\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$ is :-

(1) $\hat{i} + \hat{j}$

(2) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

(3) $\hat{j} + \hat{k}$

(4) $\frac{-\hat{i} + \hat{k}}{\sqrt{2}}$

45) The resultant of 2 forces acting at an angle of 150° is 10N and is perpendicular to one of the force. The smaller force is :-

(1) $10\sqrt{3}$N

(2) $20\sqrt{3}$N

(3) 20 N

(4) $\frac{20}{\sqrt{3}}$N

## CHEMISTRY

1)

If moving proton, dutron and alpha particle have same De-Broglie wave length then which is correct

order of their kinetic energy :-

(1) $E_p < E_\alpha < E_d$

(2) $E_d < E_p < E_\alpha$

(3) $E_p < E_d < E_\alpha$

(4) $E_\alpha < E_d < E_p$

2)

Fe shows an oxidation state of +1 in :-

(1) $Fe(CO)_5$

(2) $[Fe(H_2O)_5 NO]SO_4$

(3) $Fe_4[Fe(CN)_6]_3$

(4) $FeCl_4$

3)

In the following reaction :
$As_2S_3 + H^+ + NO_3^- \rightarrow NO + H_2O + AsO_4^{-3} + SO_4^{-2}$
The equivalent weight of $As_2S_3$ – [M = molecular weight]

(1) M/2

(2) M/4

(3) M/24

(4) M/28

4)

The incorrect order of decreasing oxidation number of S in compounds is :-

(1) $H_2S_2O_7 > Na_2S_4O_6 > Na_2S_2O_3 > S_8$

(2) $H_2SO_5 > H_2SO_3 > SCl_2 > H_2S$

(3) $SO_3 > SO_2 > H_2S > S_8$

(4) $H_2SO_0 > SO_2 > H_2S > H_2S_2O_8$

5)

Which is correct statement regarding $CrO_5$
A : Oxidation number of Cr is +6.
B : Four O-atoms of molecule are involved in peroxide linkage.
C : Only one O-atom has O.N. = –2

(1) All are correct

(2) B, C are correct

(3) A, C are correct

(4) Only B is correct

6)

Equivalent weight of $FeS_2$ in the half reaction $FeS_2 \rightarrow Fe_2O_3 + SO_2$ is :-

(1) M/10
(2) M/11
(3) M/6
(4) M/1

7) The oxidation state of molybdenum in its complex, $[Mo_2O_4(C_2H_4)_2(H_2O)_2]^{2-}$ is –

(1) +2
(2) +3
(3) +4
(4) +5

8) In the reaction, $Cl_2 + OH^- \rightarrow Cl^- + ClO_4^- + H_2O$, chlorine is :

(1) Oxidised
(2) Reduced
(3) Oxidised as well as reduced
(4) Neither oxidised nor reduced

9) How many moles of stannous oxalate can be oxidised into $Sn^{+4}$ and $CO_2$ using 1 mole of $K_2Cr_2O_7$ ?:-

(1) 0.5
(2) 1
(3) 1.5
(4) 2

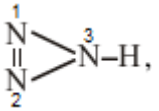
10) In a biological reaction glucose ($C_6H_{12}O_6$) is converted into 2 molecules of pyruvate ($C_3H_4O_3$). Equivalent weight of glucose in this reaction is-

(1) 180
(2) 90
(3) 45
(4) 60

11) When $CrI_3$ oxidises to $Cr_2O_7^{-2}$ and $IO_4^-$ , equivalent mass of $CrI_3$ will be :-

(1) $\frac{M}{33}$
(2) $\frac{M}{27}$
(3) $\frac{M}{28}$

(4) $\frac{M}{24}$

12) Redox reaction are those in which oxidation and reduction take place simultaneously. Oxidisin agent can gain electron whereas reducing agent can lose electron easily. The oxidation state of any element can never be in fraction. If oxidation number of any element comes out be in fraction, it is average oxidation number of that element which is present in different oxidation state.  , In this compound $HN_3$ (hydrazoic acid), oxidation state of $N^1$, $N^2$ and $N^3$ are :-

(1) 0, 0, 3
(2) 0, 0, –1
(3) 1, 1, –3
(4) –3, –3, –3

13) A 1.0 g sample of $H_2O_2$ solution contains x% by mass and requires x mL of $KMnO_4$ in acidic solution for complete oxidation. The normality of $KMnO_4$ :-

(1) 0.58
(2) 0.116
(3) 0.25
(4) 0.3

14) In an experiment, 50 mL of 0.1 M solution of a metal salt reacted with 25 mL of 0.1 M solution of sodium sulphite. The half equation for the oxidation of sulphite ion is $SO_3^{2-}(aq) + H_2O(\ell) \rightarrow SO_4^{2-}(aq) + 2H^+(aq) + 2e^-$ If the oxidation number of metal in the salt was 3, what would be the new oxidation number of metal ?

(1) Zero
(2) 1
(3) 2
(4) 4

15) $H_2O_2$ is oxidised by $MnO_4^-$ in acidic medium to form $O_2$ and $Mn^{2+}$. How many grams of $O_2$ are produced from $1.5\times10^{-3}$ moles of $MnO_4^-$ and 1.2 milligrams of $H_2O_2$ :-

(1) $1.13 \times 10^{-3}$
(2) $2.5 \times 10^{-3}$
(3) $7.5 \times 10^{-4}$
(4) $0.35 \times 10^{-5}$

16)

Incorrect equivalent mass of reactant in following column ? (M = molar mass of reactant):-

| | Column-I | Column-II |
|---|---|---|
| (1) | $FeS_2 \rightarrow Fe^{3+} + SO_3$ | $\frac{11M}{10}$ |
| (2) | $P_2H_4 \rightarrow P_4H_2 + PH_3$ | $\frac{5M}{6}$ |
| (3) | $2Mn^{+7} \rightarrow Mn^{2+} + Mn^{+6}$ | $\frac{M}{3}$ |
| (4) | $Ba(MnO_4)_2 \xrightarrow{H^+} Mn^{2+}$ | $\frac{M}{5}$ |

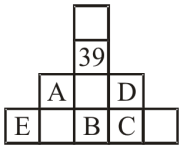
(1) A
(2) B
(3) C
(4) D

17) Be and Al show diagonal relationship hence, both have :-

(1) Almost same of electronegativity
(2) Amphoteric nature of oxides
(3) Approximately same polarizing power of respective cations
(4) All the above properties.

18) The correct order of metallic behaviour :-

(1) Na > Mg > Si > P > Be
(2) Na > Mg > Be > Si > P
(3) Be > Na > Mg > Si > P
(4) P > Be > Si > Na > Mg

19) 

On the basis of given part of periodic table, incorrect statement is :-

(1) A is an alkaline earth metal
(2) Atomic number of B is 103 which belongs to III B group
(3) Atomic number, group no. and period number of D are 72 IVB and 6th respectively
(4) C is a transuranic element.

20) IUPAC name of the element placed just after actinide series :

(1) Unniltrium
(2) Unnilpentium
(3) Unnilquadium
(4) Ununhexium

21) Correct set of Ionic Radius :-

(1) $\bar{F} > N^{-3} > Br^{-} > I^{-}$
(2) $Br^{-} > I^{-} > \bar{F} > N^{-3}$
(3) $I^{-} > Br^{-} > \bar{F} > N^{-3}$
(4) $I^{-} > Br^{-} > N^{-3} > \bar{F}$

22)

Species which are isoelectronic to one another are
(a) $CN^{-}$ (b) $OH^{-}$
(c) $CH_3^{+}$ (d) $N_2$
(e) CO

(1) a, b, c
(2) a, c, d
(3) a, d, e
(4) b, c, d

23) If the atomic number of an element is 58, it will be placed in-

(1) 5th period & 14th group
(2) 6th period & 4th group
(3) 5th period & 3rd group
(4) 6th period & 3rd group

24) What will be the distance between H and Cl atom in HCl ? If radius of hydrogen is 0.37 Å and the radius of chlorine is 1.67Å -
(According to the concept of covalent radius)

(1) 2.04 Å
(2) 1.96 Å
(3) 2.12 Å
(4) 1.0 Å

25) Which is the correct in the following ?

(1) Radius of Cl atom is 0.99Å, while that of $Cl^{+}$ ion is 1.54 Å
(2) Radius of Cl atom is 0.99Å, while that of Na atom is 1.54 Å
(3) Radius of Cl atom is 0.99Å, while that of Cl– ion is 0.81 Å
(4) Radius of Na atom is 0.95Å, while that of $Na^{+}$ ion is 1.54 Å

26) Arrange $Ce^{3+}$, $La^{3+}$, $Pm^{3+}$ and $Yb^{3+}$ in increasing order of their size:-

(1) $Yb^{3+} < Pm^{3+} < Ce^{3+} < La^{3+}$
(2) $Ce^{3+} < Yb^{3+} < Pm^{3+} < La^{3+}$

(3) $Yb^{3+} < Pm^{3+} < La^{3+} < Ce^{3+}$

(4) $Pm^{3+} < La^{3+} < Ce^{3+} < Yb^{3+}$

27) In which of the following compounds chromium shows maximum radius ?

(1) $K_2Cr_2O_7$

(2) $CrO_2Cl_2$

(3) $Cr_2(SO_4)_3$

(4) $CrCl_2$

28) If uncertainty in position and momentum are equal, then uncertainty in velocity is :

(1) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

(2) $\sqrt{\frac{h}{2\pi}}$

(3) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$

(4) $\sqrt{\frac{h}{\pi}}$

29) An electron beam can undergo diffraction by crystals which proves the wave nature of electrons. The potential required for a beam of electrons to be accelerated from rest so that its wavelength becomes equal to 0.154 nm is

(1) 63.5 V

(2) 31.75 V

(3) 635 V

(4) 127 V

30) Total number of elements of 5th period which have one or more than one 5d electrons :-

(1) 10

(2) 16

(3) 30

(4) Zero

31) If a periodic table is prepared by subsiding the rules to write electronic configuration (Except Pauli's principle). Then the no. of elements in 6th period of P.T. would be :-

(1) 32

(2) 50

(3) 72

(4) 128

32) According to Slater, $Z_{eff}$ order for Li, Na, K is :-

(1) Na > Li > K

(2) Li = Na = K

(3) Li < Na = K

(4) None

33) The set representing the correct order of ionic radius is :-

(1) $Na^{+} > Mg^{+2} > Al^{+3} > Li^{+} > Be^{+2}$

(2) $Na^{+} > Li^{+} > Mg^{+2} > Al^{+3} > Be^{+2}$

(3) $Na^{+} > Mg^{+2} > Li^{+} > Al^{+3} > Be^{+2}$

(4) None of these

34) Which of the following order of radius is incorrect ?

(1) $Ti^{+4} < Zr^{+4} \approx Hf^{+4}$

(2) Sc < Y < La

(3) $La^{+3} > Pr^{+3} > Gd^{+3} > Yd^{+3}$

(4) $F^{-} > O^{-2} > F > O$

35) The period number and group number of Ta(Z=73) are respectively :-

(1) 5, VII B

(2) 6, III A

(3) 6, V B

(4) None

36) The element having electronic configuration $[Kr]4d^{10}4f^{14}$, $5s^2$ $5p^6$, $6s^2$ belongs to :-

(1) s-block

(2) p-block

(3) d-block

(4) f-block

37)

| | Column-I (Type of element) | | Column-II (Outer electronic configuration) |
|---|---|---|---|
| (A) | Inert gas elements | (i) | $ns^{1-2}$ and $ns^2$ $np^1$ to $ns^2$ $np^5$ |
| (B) | Representative elements | (ii) | $1s^2$ and $ns^2$ $np^6$ |
| (C) | Transition elements | (iii) | $(n-2)f^{1-14}$ $(n-1)d^{0-1}$ $ns^2$ |
| (D) | Inner-transition elements | (iv) | $(n-1)d^{1-10}$ $ns^{1 \text{ or } 2}$ |

(1) A-i, B-ii, C-iii, D-iv

(2) A-ii, B-i, C-iii, D-iv

(3) A-ii, B-i, C-iv, D-iii

(4) A-i, B-ii, C-iv, D-iii

38) Vishal thakur went to meet his friend Sumit, where he saw that his friend was doing the study of a particular chemistry book. But he could not find the theoretical value of bond length in H- F but he found that $r_H$ and $r_F$ are 0.37 Å and 0.72 Å respectively and electronegativity of F and H are H are 4.0 and 2.1 respectively. What is bond length of H – F bond :-

(1) 1.09

(2) 1.784

(3) 0.92

(4) 0.46

39) Which of the following order of radius is not correct

(1) $Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

(2) $Mg^{+2} < Na^{+} < Al < F^{-}$

(3) $K > Ca > Mg > Li$

(4) $O < O^{-2} < F < F^{-}$

40) Match the column :-

| | **Column I** | | **Column II** |
|---|---|---|---|
| a | S, Se | P | Chalcogen family |
| b | Cl, Br | Q | Halogen family |
| c | N, P | R | Pnicogen family |
| d | Be, Al | S | Diagonal relationship |
| | | T | Nonmetals |

(1) [a – P,T], [b – Q,T], [c – S,T], [d – S]

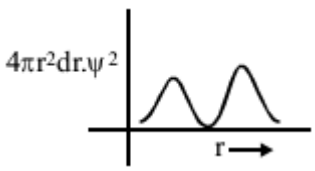
(2) [a – P,T], [b – Q,T], [c – R,T], [d – S]

(3) [a – Q,T], [b – P,T], [c – R,T], [d – S]

(4) [a – R,T], [b – P,T], [c – Q,T], [d – S]

41)

This curve can show radial probability distribution of :-



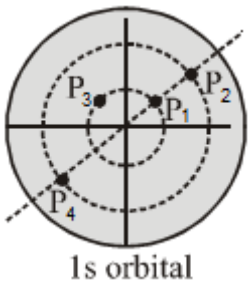

(a) 2p (b) 2s
(c) 4d (d) 3p

(1) c, d

(2) b, c, d

(3) Only b

(4) a, b

42) The given diagram shows points $P_1$, $P_2$ $P_3$ & $P_4$ in 1s orbital. 

1s orbital

The correct order of radial probability density at these point are

(1) $P_1 = P_2 = P_4 > P_3$

(2) $P_1 = P_3 < P_2 = P_4$

(3) $P_1 = P_3 > P_2 = P_4$

(4) $P_1 = P_2 = P_4 < P_3$

43) Which of the following statement is true :-
(a) 6 electrons are present in Mg for which m = 0
(b) Maximum 6 electrons can accomdate in $p_x$-orbital
(c) Maximum 18 electrons present in M-shell
(d) 3–electron present in phosphorous for which,
l = 0, S = $+\frac{1}{2}$

(1) a, b, c, d

(2) a, b c

(3) a, c, d

(4) c, d

44)

Consider the following statements :-
(a) Electron density in xy plane in $3d_{x^2-y^2}$ orbital is zero.
(b) Electron density in xy plane in $3d_{z^2}$ orbital is zero.
(c) 2s orbital has only one spherical node
(d) For $2p_z$ orbital yz is the nodal plane
The correct statements are

(1) (b) and (c)

(2) (a), (b), (c) (d)

(3) Only (c)

(4) (a) and (c)

45) Which orbital can not have $|m| = (n-2)$ ?

(1) $4d_{xy}$

(2) $3p_{y}$

(3) 2S

(4) None of these

## BIOLOGY

1) A great diversity is there in different organisms but in taxonomic hierarchy at any level they prove their linkage to each other. Wheat and mango also show such type of linkage, but at which level?

(1) Family

(2) Class

(3) Division

(4) Order

2) Unicellular organisms shows growth and reproduction through :-

(1) Cell division

(2) Sexual reproduction

(3) Only binary fission

(4) Vegetative propagation

3) Consider the following statements. Which of the following statements are related to universal rules of nomenclature ?
(a) Biological names are generally in Latin.
(b) The first word in a biological name represents the generic name while the second component denotes the specific epithet.
(c) Both the words in a biological name, when printed, are separately underlined, to indicate their latin origin.
(d) Generic name starts with a capital letter while the specific epithet starts with a small letter.

(1) Only b, c & d

(2) Only c & d

(3) Only a, c & d

(4) Only a, b & d

4) According to carl woese classification of wheat plant is classified in to :-

(1) 5 - Taxonomic Categories

(2) 6 - Taxonomic Categories

(3) 7 - Taxonomic Categories

(4) 8 - Taxonomic Categories

5) *Mangifera indica, Saraca indica, Rosa indica, Tamarindus indicus, Solanum tuberosum, Solanum melongena* and *Solanum nigrum*.
These are some names of plants. How many species and genera are included respectively in these

names :-

(1) 7, 5
(2) 5, 7
(3) 7, 7
(4) 5, 5

6) The species (Man, Housefly, Mango, Wheat, Dog, Cat, Lion, Tiger, Potato, Brinjal, Makoi and Leopard) given here belong to how many different families ?

(1) 4
(2) 7
(3) 5
(4) 6

7) Read the following statements :-
A. Isolated-metabolic reactions in-vitro are neither living nor non-living.
B. Reproduction is synonymous with growth in Chlamydomonas.
C. Reproduction is an all inclusive defining characteristic of living organisms.
D. Extrinsic growth cannot be taken as defining property of living organisms.
How many of the above statement(s) is/are not true ?

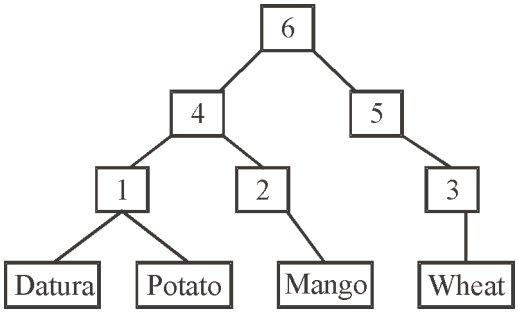
(1) One
(2) Two
(3) Three
(4) Four

8)

Among the following, select the **correct** statements:-
(a) In majority of higher plants and animals, growth and reproduction are mutually exclusive events.
(b) In non-living objects growth is by accumulation of material on the surface.
(c) An isolated metabolic reaction outside the body of an organism, performed in a test tube is neither living nor non-living.
(d) All organisms, from the prokaryotes to the most complex eukaryotes can sense and respond to environmental cues.

(1) a, b, c
(2) a, c, d
(3) a, b, d
(4) a, b, c, d

9) Recognise the following flow diagram and find out the correct option according to taxonomic

hierarchy. 


(1) 1-Polymoniales, 2-Sapindales, 3-Poales, 4-Dicotyledonae, 5- Monocotyledonae, 6-Angiospermae

(2) 1-Sapindales, 2-Polymoniales, 3-Poales, 4-Monocotyledonae, 5-Dicotyledonae, 6-Angiospermae

(3) 1-Polymoniales, 2-Sapindales, 3-Poaceae, 4-Dicotyledonae, 5-Monocotyledonae, 6-Gymnospermae

(4) 1-*Solanum*, 2-*Mangifera*, 3-*Triticum*, 4-Angiospermae, 5-Monocotyledonae, 6-Angiospermae

10) Assertion A: Scientific names of organisms are unique and universally accepted.
Reason R: International Code for Botanical Nomenclature (ICBN) and International Code of Zoological Nomenclature (ICZN) provide principles and criteria for naming plants and animals respectively.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

11) **Assertion** :- Genera are aggregates of closely related species. **Reason** :- Genus comprises a group of related species which has more common characters in comparison to species of other genera.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(2) Assertion is True but the Reason is False.

(3) Assertion is False but the Reason is True.

(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

12) Read the following statement(A-D) & select the option which include all the correct ones only.
(A) In majority of higher animals & plants growth and reproduction are mutually exclusive events.
(B) In living organisms growth is from inside.
(C) The sum total of all the chemical reactions occurring in our body is Anabolism.
(D) In plants, growth by cell division occurs continuously throughout their life span.

(1) Statements A, C, D

(2) Statements A, B, D

(3) Statements A, B, C

(4) Statements B, C, D

13) Consider the following four statements (a-d) and select the option which includes all the correct ones only :-
(a) Photoperiod affects reproduction in seasonal breeders, both plants and animals.
(b) In unicellular organisms growth and reproduction are mutually exclusive events.
(c) Metabolic reactions can be made to perform outside the body in a test tube.
(d) Cellular organisation is absent in many living organisms

(1) Statements b, d

(2) Statements a, c

(3) Statements a, b, c

(4) Statements a, b, d

14) Read the following statements and select the incorrect one :-

(1) Isolated metabolic reactions in vitro are not living things but surely living reactions.

(2) Some organisms reproduce by both asexual and sexual methods.

(3) All organisms from prokaryotes to the most complex eukaryotes have self consciousness.

(4) Metabolic reactions can be demonstrated outside the body in cell free system.

15) Assertion A: Taxonomy involves the processes of characterisation, identification, classification, and nomenclature.
Reason R: These processes are based on the external and internal structures, cell structure, development processes, and ecological information of organisms.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

16) Which of the following features are generally characteristics of life ?
(I) Growth
(II) Reproduction
(III) Response to stimuli
(IV) Metabolism
(V) Cellular organization

(1) Only I, II, IV, V

(2) Only I, II, III, V

(3) Only II, III, IV

(4) All of these

17)

Study the four statements (A–D) given below and select the two **correct** ones out of them :
A. Definition of biological species was given by Ernst Mayr.

B. Photoperiod does not affect reproduction in plants.
C. Binomial nomenclature system was given by R.H. Whittaker.
D. In unicellular organisms, reproduction is synonymous with growth.
The two correct statements are

(1) A and D

(2) A and B

(3) B and C

(4) C and D

18) Among the following, identify the properties shown by the living organisms exclusively and choose the correct option from the codes given below.
**I.** Internal Growth
**II.** Increase in mass
**III.** Self-replication ability
**IV.** Response to touch stimuli
**Codes**

(1) I and II

(2) I, II and III

(3) I, III and IV

(4) I, II, III and IV

19) Match the following columns.

| | **Column I** | | **Column II** |
|---|---|---|---|
| **A.** | *Planaria* | **1** | Binary fission |
| **B.** | Fungi | **2** | Asexual spores |
| **C.** | Yeast | **3** | Budding |
| **D.** | *Ulothrix* | **4** | True regeneration |
| **E.** | *Amoeba* | **5** | Fragmentation |

(1) A – 1, B – 2, C – 4, D – 5, E – 3

(2) A – 4, B – 2, C – 3, D – 5, E – 1

(3) A – 5, B – 2, C – 3, D – 1, E – 4

(4) A – 4, B – 5, C – 1, D – 3, E – 2

20)

The correct sequential steps of taxonomy for a newly discovered organism is :

(1) Description/Characterisation → Nomenclature → Identification → Classification

(2) Classification → Nomenclature → Description → Identification

(3) Identification → Nomenclature → Description → Classification

(4) Description/Characterisation → Identification → Nomenclature → Classification

21) A. The first word denoting the genus start with a small letter, while second word is specific start with capital letter.
B. Biological names are generally printed in italics.

In given statement, which of the following is incorrect and correct.

(1) A-is incorrect and B-is correct
(2) A-is correct and B-is incorrect
(3) A and B both are incorrect
(4) A and B both are correct

22) Which of the following statements regarding growth is incorrect?

(1) In plants, growth by cell division is seen only upto a certain stage.
(2) Growth exhibited by non-living objects is by accumulation of material on the surface.
(3) A multicellular organism grows by cell division.
(4) Growth in *in–vitro* culture of unicellular organisms can be observed by counting the number of cells.

23) All the below given phenomena exemplify consciousness in living beings, **except**

(1) Flowering in plants in specific season
(2) Reproduction of goats in specific season
(3) Stomatal closure at high temperature
(4) Breakdown of sugar into water and $CO_2$

24) State **True** (T) or **False** (F) to the following statements and choose the correct option,
A. All organisms from prokaryotes to the complex eukaryotes can sense various changes in their surroundings.
B. All properties of tissues are present in their constituent cells,
C. Consciousness is the most obvious and technically complicated feature of all living organisms.
D. Increase in mass and increase in number of cells are twin characteristics of growth in living organism.

| | A | B | C | D |
|---|---|---|---|---|
| (1) | T | F | T | F |
| (2) | F | T | T | F |
| (3) | T | F | T | T |
| (4) | T | T | F | F |

(1) 1
(2) 2
(3) 3
(4) 4

25) Which of the following can grow but cannot reproduce?

(1) Single celled organisms
(2) Living organisms
(3) Non-livings only

(4) Both (1) & (3)

26) Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
**Assertion (A) :** Living organisms shows similarities both horizontally and vertically.
**Reason (R) :** All present day living organisms are related to each other and also to all organisms that ever lived on this earth; we all originate from universal common ancestor - Progenote.
In the light of the above statements, choose the correct answer from the options given below :

(1) (A) is correct but (R) is not correct

(2) (A) is not correct but (R) is correct

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

27) The two events which are mutually inclusive in unicellular organisms are :-

(1) Reproductive and metabolism

(2) Cellular organization and consciousness

(3) Sensitivity and self-consciousness

(4) Growth and reproduction.

28) Which of the following feature given below are exhibited by bacteria (A) and higher plants (B)?
(i) Grow and reproduction both are synonyms
(ii) Binary fission
(iii) Cellular organization
(iv) Ability to sense their environment

| | **A** | **B** |
|---|---|---|
| (1) | (i), (ii) | (ii), (iii), (iv) |
| (2) | (i), (iii), (iv) | (i), (ii) |
| (3) | (ii), (ii) | (i), (iii), (iv) |
| (4) | (i), (ii), (iii), (iv) | (iii), (iv) |

(1) 1

(2) 2

(3) 3

(4) 4

29) Find out the **incorrect** statements:
(a) Isolated metabolic reactions can be demon- strated *in vitro* are not living things
(b) All living organisms has ability to sense their surroundings
(c) Photoperiod affects reproduction in seasonal and non seasonal breeders

(1) a, b,c

(2) Only c

(3) Only b, c

(4) Only a, d

30) Read the statements A and B and select the correct option.
A. Patient lying in coma has no self-consciousness.
B. Patient lying in coma is brain dead and therefore considered as non- living object.

(1) Only A is correct
(2) Only B is correct
(3) Both A and B are correct
(4) Both A and B are incorrect

31) Read the following statement A and B and choose the correct option :
**Statement A :-** The number of common characteristics goes on increasing from species to kingdom
**Statement B :-** Higher the category, greater is the difficulty of determining the relationship to other taxa of same level.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

32) **Assertion :-** Increase in body mass is considered as growth.
**Reason :-** Increase in number and increase in mass of individuals are twin characters of growth.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

33) **Assertion :** Insects represents a group of organisms sharing common features like three pairs of jointed legs.
**Reason :** Taxonomic groups are distinct biological entities and not merely morphological aggregates.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

34) **Assertion :** Potato and brinjal are two different species but both belong to the genus *Solanum*.
**Reason :** Genus comprises a group of related species which has more characters in common in comparison to species of other genera.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

35) **Assertion :** Animals, mammals and dogs represent taxa at different levels.
**Reason :** Genus is the smallest unit of taxonomy

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

36) Technically living organisms are :

(1) Self organising, self regulating dead interactive system.
(2) Self organising, self regulating closed interactive system.
(3) Self organising, self regulating interactive system capable of responding to external stimuli
(4) Self organising, inactive system.

37) **Statement-I :** In majority of higher animal and plants growth and reproduction are mutually exclusive events.
**Statement-II :** Non-living object also grow if we take increase in body mass as a criterion for growth.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II is incorrect.

38) **Statement - I :** Beside growth, cell division occurs in certain tissues to replace lost cells.
**Statement - II :** Growth can be taken as a defining property of living organism.

(1) Statement I and II both are correct.
(2) Statement I and II both are incorrect.
(3) Only Statement I is correct.
(4) Only Statement II is correct.

39) Read the following statements and select the incorrect one :-

(1) Mountains, boulders and sand dunes do grow if we take increase in body mass as criterion for growth.
(2) Many organisms like mules, sterile worker bees and infertile human couples donot reproduce at all.
(3) Living organism are self-replicating, evolving and self-regulating interactive systems capable of responding to external stimuli.
(4) Isolated metabolic reactions *invitro* are living beings.

40) Choose the correct sequence of taxonomic categories in descending order

(1) Phylum - Class - Order - Family - Genus
(2) Kingdom - Division - Class - Family - Order

(3) Genus - Family - Order - Class - Division

(4) Kingdom - Phylum - Order - Species - Genus

41) Study the following table which shows different organisms with their taxonomic categories -

| Common name | Family | Order | Class | Phylum/Division |
|---|---|---|---|---|
| Man | Hominidae | Primata | Mammalia | A |
| Houselfy | Muscidae | Diptera | B | Arthropoda |
| Mango | C | Sapindales | Dicotyledonae | Angiospermae |
| Wheat | Poaceae | Poales | D | Angiospermae |

Select the correct option for A, B, C and D

(1) A-Chordata, B-Insecta, C-Anacardiaceae, D-Monocotyledonae

(2) A-Animalia, B-Anacardiaceae, C-Solanaceae, D-Dicotyledonae

(3) A-Chordata, B-Anacardiaceae, C-Polymoniales, D-Monocotyledonae

(4) A-Non-chordata, B-Insecta, C-Anacardiaceae, D-Dicotyledonae

42) How many features listed below exhibited by bacteria.
i. Growth by increase in mass only
ii. Regeneration
iii. Consciousness
iv. Metabolism
v. Binary fission

(1) iii, iv, v

(2) i, ii, iii

(3) ii, iv, v

(4) i, iv, v

43) A. In plant, growth generally indeterminate.
B. Growth measured by a variety of parameters like weight, length, area, volume and cell number.
In given

(1) A is correct, B is wrong

(2) A is wrong, B is correct

(3) A and B both are wrong

(4) A and B both are correct

44)

How many statements are not incorrect -

(I) Protoplasmic growth is defining property.
(II) Growth in non-living occurs by increase in number.
(III) Reproduction is most obvious and complicated process and occurs in all.
(IV) Cells contain same properties as present in tissue.

(1) Only (I) is correct

(2) (II) & (III) are correct

(3) (III) & (IV) are correct

(4) Only (IV) is correct

45) (A) Properties of tissues are present in constituent cell & do not arise as a result of interactions among constituent cells.
(B) Properties of cellular organelles are not present in the molecular constituents of the organalles but arise as a result of interaction among the molecular components comprising the organelle

(1) A – Correct ; B – Correct

(2) A – Incorrect ; B – Correct

(3) A – Correct ; B – Incorrect

(4) A – Correct ; B – Incorrect

46) Read the following statements carefully and choose the option with all the correct ones only:-
(A) Total thickness of diffusion membrane is much less than a millimetre.
(B) Solubility of gases does not affect the rate of diffusion.
(C) Partial pressure of $CO_2$ in tissues is lower than the oxygenated blood.
(D) The diffusion membrane is made up of three major layers.
Options :-

(1) A, B and C

(2) A and D

(3) B and C

(4) B, C and D

47) The correct match for kind of respiration is :-

| **Column I Animal** | | **Column II Respiration** | |
|---|---|---|---|
| (A) | Earthworm | (I) | Pulmonary |
| (B) | Human | (II) | Branchial |
| (C) | Prawn | (III) | Tracheal |
| (D) | Insects | (IV) | Cutaneous |

(1) A – I, B – II, C – III, D – IV

(2) A – IV, B – II, C – I, D – III

(3) A – IV, B – I, C – II, D – III

(4) A – III, B – II, C – IV, D – I

48) Which of the following is not a correctly matched pair :-
(A) Inspiratory reserve volume – 2500-3000 ml
(B) Residual volume – 1500-1600 ml
(C) Expiratory capacity – 1000-1100 ml
(D) Functional residual capacity – 2200-2300 ml

(1) A, D

(2) B, C

(3) A, C

(4) B, C, D

49) Identify the correct and incorrect match about the respiratory volume and capacities and mark the correct answer :-
(a) Expiratory capacity = Tidal volume + Residual volume
(b) Vital capacity = Total lung capacity – Residual volume
(c) Inspiratory capacity = Total lung capacity – Functional Residual capacity
(d) Total lung capacity = Vital capacity + Functional Residual capacity

| | **(a)** | **(b)** | **(c)** | **(d)** |
|---|---|---|---|---|
| (1) | Incorrect | Correct | Correct | Incorrect |
| (2) | Correct | Correct | Incorrect | Correct |
| (3) | Incorrect | Correct | Incorrect | Incorrect |
| (4) | Correct | Incorrect | Incorrect | Correct |

(1) 1

(2) 2

(3) 3

(4) 4

50) **Assertion :** The formation of Bicarbonate 70% takes place in the RBC not in the plasma.
**Reason :** RBCs contain very high concentration of the enzyme carbonic anyhydrase.

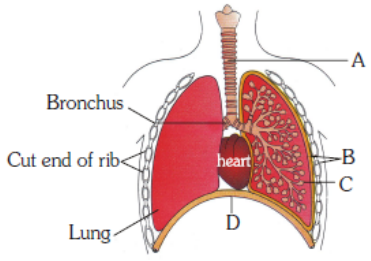
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion are True but Reason is not correct.

(4) Assertion is false but Reason is true.

51) The figure shows a diagrammatic view of human respiratory system with labels A, B, C and D. Select the option which gives correct identification and main function and/or characteristic :-



(1) D – Lower end of lungs – diaphragm pulls it down during inspiration

(2) A – trachea - long tube supported by complete cartilaginous rings for conducting inspired air

(3) B – pleural membrane - surround ribs on both sides to provide cushion against rubbing

(4) C – Alveoli - thin walled vascular bag like structures for exchange of gases

52) Which of the following values of volumes and capacities related with respiratory system cannot be measured by spirometer.

(1) VC

(2) IRV

(3) ERV

(4) RV

53) Glottis is an opening in the floor of :-

(1) Mouth

(2) Trachea

(3) Pharynx

(4) Diaphragm

54) Lining of trachea is made up of

(1) stratified ciliated epithelium

(2) pseudostratified ciliated epithelium

(3) simple squamous epithelium

(4) stratified cuboidal epithelium

55) **Assertion :** Gaseous exchanges continue in the lungs without interruption even during expiration.
**Reason :** Residual volume of air can never be driven out by respiration.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

56) Vocal cords occur in -

(1) Larynx

(2) Pharynx

(3) Glottis

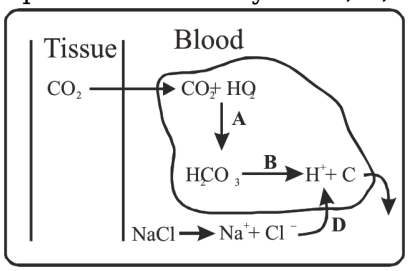
(4) Bronchial tube

57) If A = Primary bronchi, B = Alveolar duct,
C = Alveoli, D = Secondary bronchi,
E = Terminal

bronchiole, then starting from trachea (T), which sequence of branching is correct?

(1) T → A → E → D → C → B
(2) T → A → D → E → B → C
(3) T → A → D → E → C → B
(4) T → E → A → D → B → C

58) The given diagram represent the $CO_2$ transportation. Identify the A, B, C and phenomena D in this diagram and choose the correct option :-




(1) A - Carbonic anhydrase, B - Carbonic anhydrase, C - $HCO_3$– and D - haldane effect
(2) A - Carbonic anhydrase, B - Carbonic anhydrase, C - $HCO_3$– and D - Hamburger phenomenon
(3) A - Carbonic anhydrase, B - Carbonic anhydrase, C - $HCO_2$– and D - Chloride phenomena
(4) A - Carbonic anhydrase, B - Carbonic anhydrase, C - $HCO_2$– and D - Bohr phenomena

59) Which can activate the chemosensitive area situated adjacent to the rhythm center :-

(1) High $CO_2$, less $H^+$
(2) High $CO_2$, high $H^+$
(3) Less $CO_2$, Less $H^+$
(4) Less $CO_2$, Less $H^+$

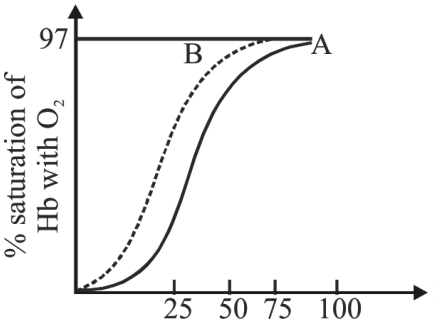
60) ***Assertion :-*** Mammals can breathe and eat at the same time.
***Reason :-*** Mammals have negative breathing pressure.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

61) Which of the following would have same $CO_2$ content ?

(1) Blood entering the lungs and leaving the lungs
(2) Blood entering in tissue and leaving the tissue
(3) Blood entering right side and leaving left side of heart
(4) Blood entering right side of the heart and leaving right side of the heart

62) On accompanying graph, the shift from curve A to B could be caused by :-




(a) Increase in pH (b) Lower Temp
(c) Increased altitude (d) Low $CO_2$

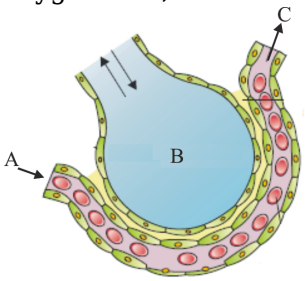
(1) a only
(2) a and d only
(3) a, b, d
(4) All

63)

How many molecule of $O_2$ binds with each $Fe^{++}$ of haemoglobin :-

(1) 4
(2) 1
(3) 2
(4) 1.34

64) Study the following diagram and select the option that correctly identifies the partial pressure of oxygen at A, B and C ?




| | **A** | **B** | **C** |
|---|---|---|---|
| (1) 40 mm Hg | 104 mm Hg | 95 mm Hg | |
| (2) 95 mm Hg | 104 mm Hg | 40 mm Hg | |
| (3) 40 mm Hg | 95 mm Hg | 104 mm Hg | |
| (4) 40 mm Hg | 45 mm Hg | 95 mm Hg | |

(1) 1
(2) 2
(3) 3
(4) 4

65) ***Assertion :-*** Strong pneumotaxic signal results complete filling of lungs.
***Reason :-*** Strong pneumotaxic signal increases the duration of inspiration as well as expiration.

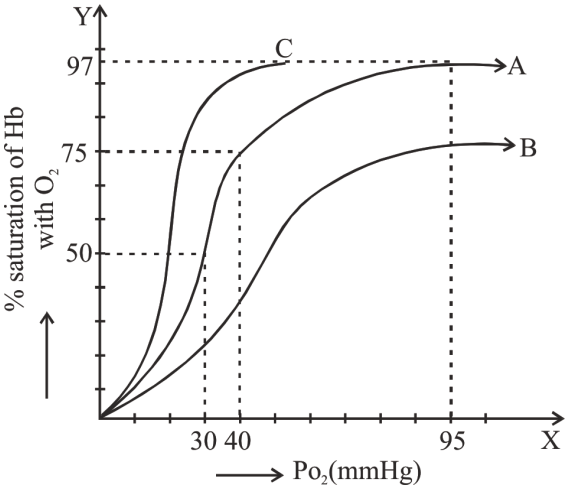
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

66)

Given below the three oxyhaemoglobin dissociation curves A, B and C plotted at different condition of body temperature :-




Find out the correct answer with regards to different ODC **Options** :-

(1) Curve 'A' is plotted at 30°C,
'B' is plotted at normal body temperature,
'C' is plotted at 40°C,

(2) Curve 'A' is plotted at 37°C,
'B' is plotted at 40°C,
'C' is plotted at 33°C,

(3) Curve 'A' is plotted at 40°C,
'B' is plotted at 45°C,
'C' is plotted at 47°C,

(4) Curve 'A' is plotted at normal body temperature
'B' is plotted at low body temperature
'C' is plotted at high body temperature

(1) 1

(2) 2

(3) 3

(4) 4

67) The thoracic chamber is formed dorsally by the A , ventrally by the B , laterally by the C and on the lower side by the dome-shaped D .

(1) A-Sternum

(2) B-Diaphragm

(3) C-Ribs

(4) D-Vertebral column

68) Read the following statements.

(a) High $H^{\oplus}$ concentration and higher temperature are favourable for dissociation of oxygen from oxyhaemoglobin .

(b) Partial pressure of oxygen in alveoli is higher than in Tissue.
(c) The maximum volume of air a person can breathe in after a forced expiration is vital capacity
(d) In alveoli $PCO_2$ low and $PO_2$ high are for dissociation of $CO_2$ from carbamino-haemoglobin.
How many statements are correct?

(1) One
(2) Two
(3) Three
(4) Four

69) Under normal condition 100 ml oxygenated blood deliver_____to tissues :-

(1) 5 ml $O_2$
(2) 10 ml $O_2$
(3) 1.34 ml $O_2$
(4) 30 ml $O_2$

70) ***Assertion :-*** Larynx is a cartilaginous box which helps in sound production.
***Reason :-*** Epiglottis flap covers the glottis during swallowing and prevents the entry of food into the larynx.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(2) Assertion is true but Reason is false.
(3) Assertion is false but Reason is true.
(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

71) People who have migrated from the planes to an area adjoining Rohtang pass about six months back

(1) Suffer from altitude sickness with symptoms like nausea, fatigue etc.
(2) Have the usual RBCs count but their haemoglobin has very high binding affinity to $O_2$.
(3) Have more RBCs and their haemoglobin has a lower binding affinity to $O_2$.
(4) Are not physically fit to play games like football

72)

The conducting part of respiratory system is responsible for :-
A. The transport of atmospheric air to alveoli
B. Clears air from foreign particles
C. Humidifies air

D. Brings the air to body temperature
(1) Only A and C
(2) Only A,B and C
(3) Only A, B and D
(4) All A,B,C and D

73) If total $O_2$ transported by blood from lungs to cell in a minute is 500 ml. Then the volume of $O_2$ transported by blood found dissolved in plasma is

(1) 970 ml

(2) 15 ml

(3) 1000 ml

(4) 30 ml

74) How much $CO_2$ is delivered to the alveoli by every 200 ml of deoxygenated blood ?

(1) 4 ml

(2) 8 ml

(3) 12 ml

(4) 3 ml

75) Some statements are given, read them and choose disorder :-
(i) Allergic condition
(ii) Difficulty in breathing
(iii) Increased $I_gE$
(iv) Wheezing sound

(1) Emphysema

(2) Anthracis

(3) Silicosis

(4) Asthma

76) Select the incorrectly matched pair :-

(1) Additional volume of air a person can expire by a forceful expiration – Expiratory reserve volume

(2) Maximum amount of air expelled forcefully from lungs after maximum forceful inspiration – Vital capacity

(3) Volume of air inspired or expired during a normal respiration – Tidal volume

(4) When a person breathes normally, the amount of air which remains in the lungs after normal expiration – Residual volume

77) When under certain conditions, the P50 value of haemoglobin rises, the affinity of the pigment of combining with oxygen will :-

(1) Remain same

(2) Rise

(3) Fall

(4) First fall and then rise

78) Pneumotaxic center is situated in :-

(1) Medulla oblongata

(2) Cerebellum

(3) Pons

(4) Thalamus

79) Proliferation of fibrous tissue in lungs of stone mine workers is characterstic feature of :-

(1) Asthma

(2) Occupational Respiratory Disorder

(3) Emphysema

(4) Pleurisy

80) The contribution of diaphragm in breathing?

(1) 25–30%

(2) 75–80%

(3) 35–45%

(4) 50%

81) **Assertion :** $CO_2$ diffuses only from tissue to alveoli and not in reverse direction.
**Reason :** $CO_2$ is 10 times more soluble than $O_2$.

(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion

(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) If Assertion is True but the Reason is False.

(4) If both Assertion & Reason are False.

82) Read the following steps of inspiration. (a) Increases thoracic volume.
(b) Air move into lungs
(c) Contraction in diaphragm and EICM
(d) Increases pulmonary volume
(e) Lungs expand
(f) Decreases the intra pulmonary pressure (IPP)
Find out the **correct** sequence of these steps.

(1) a, b, c, d, e, f

(2) c, d, a, e, f, b

(3) c, a, e, d, f, b

(4) c, e, d, a, f, b

83) Chronic disorder in which alveolar walls are damaged which leads to decrease in the respiratory surface.

(1) Asthma

(2) Emphysema

(3) Occupational respiratory disorder

(4) None of the above

84) Arrange the following in the order of increasing volume -
1. Tidal volume
2. Residual volume
3. Expiratory reserve volume
4. Inspiratory reserve volume

(1) 1 < 2 < 3 < 4
(2) 1 < 4 < 3 < 2
(3) 1 < 3 < 2 < 4
(4) 1 < 4 < 2 < 3

85) Reduction in pH of blood will

(1) Release bicarbonate ions by the liver
(2) Reduce the rate of heart beat
(3) Reduce the blood supply to the brain
(4) Decrease the affinity of hemoglobin with oxygen

86) If a person exhales out forcefully by applying all his efforts. What will the pulmonary volume inhaled by him immediately under normal condition without applying any extra effort?

(1) TV + IRV
(2) TV + RV
(3) TV + ERV
(4) TV + IRV + ERV

87) Which one of the following mammalian cells is not capable of metabolising glucose to carbon-dioxide aerobically?

(1) Red blood cells
(2) White blood cells
(3) Unstriated muscle cells
(4) Liver cells

88) Habit of Cigarette smoking can lead to :

(1) loss of cilia lining the respiratory tract
(2) emphysema
(3) coughing
(4) All of the Above

89) If TLC is 5500ml, IRV is 2950ml, ERV is 900ml and TV is 500ml then what will be value of RV?

(1) 2550ml
(2) 1100ml
(3) 1200ml

(4) 1150ml

90) **Assertion :** It is evident that oxygen has to be continuously provided to the cells and carbon dioxide produced by the cells have to be released out.
**Reason :** Oxygen is utilised by the organism to indirectly breakdown of glucose to derive energy for performing various activities and carbon dioxide which is harmful is also released during the above catabolic reactions.

(1) Both Assertion and Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion and Reason are False.

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 1 | 3 | 3 | 1 | 4 | 2 | 4 | 2 | 1 | 4 | 1 | 2 | 4 | 4 | 1 | 1 | 4 | 1 | 1 | 3 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 3 | 3 | 2 | 2 | 1 | 1 | 4 | 3 | 3 | 2 | 2 | 3 | 2 | 1 | 4 | 2 | 1 | 3 | 2 | 1 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 3 | 2 | 2 | 4 | 1 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 4 | 2 | 4 | 4 | 1 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 1 | 3 | 1 | C | 4 | 2 | 2 | 3 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 4 | 3 | 4 | 2 | 2 | 1 | 4 | 1 | 1 | 4 | 3 | 3 | 2 | 4 | 3 | 4 | 3 | 3 | 4 | 2 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 2 | 3 | 4 | 3 | 4 | | | | | | | | | | | | | | | |

## BIOLOGY

| Q. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 1 | 4 | 4 | 1 | 2 | 1 | 4 | 1 | 1 | 4 | 2 | 2 | 3 | 2 | 4 | 1 | 3 | 2 | 4 |
| Q. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| A. | 1 | 1 | 4 | 3 | 3 | 3 | 4 | 4 | 2 | 1 | 2 | 1 | 2 | 1 | 3 | 3 | 1 | 3 | 4 | 1 |
| Q. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| A. | 1 | 1 | 4 | 1 | 2 | 2 | 3 | 2 | 1 | 1 | 4 | 4 | 3 | 2 | 1 | 1 | 2 | 2 | 2 | 4 |
| Q. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| A. | 4 | 3 | 2 | 1 | 4 | 2 | 3 | 4 | 1 | 1 | 3 | 4 | 2 | 2 | 4 | 4 | 3 | 3 | 2 | 2 |
| Q. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | | | | | | | | | | |
| A. | 3 | 3 | 2 | 3 | 4 | 3 | 1 | 4 | 4 | 1 | | | | | | | | | | |

SOLUTIONS

PHYSICS

1) $F = \frac{dm}{dt} \times v = 0.5 \times 2 = 1N$

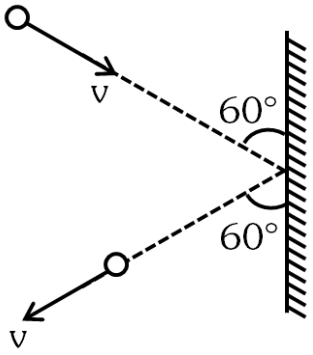
2) Let natural length of spring be $\ell_0$, then according to question,

$4 = k(a - \ell_0)$ and $5 = k(b - \ell_0)$

$\Rightarrow \quad l^0 = 5a - 4b$ and $k = \frac{1}{b-a}$

Now if we apply 9 N force, then

$9 = k(\ell - \ell_0) \Rightarrow \ell = 5b - 4a$



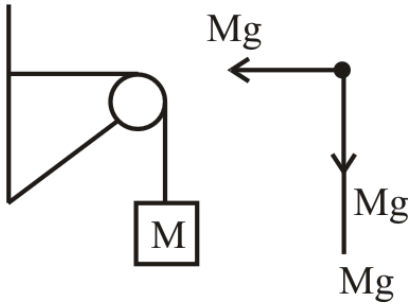

3) $F = \left|\frac{\Delta\vec{p}}{\Delta t}\right| = \frac{2mv\sin\theta}{t} = \frac{2(1)(1)\sin 60^\circ}{0.1} = 10\sqrt{3}\,N$

4) $Mg \times \frac{1}{2} = mg$

$\frac{M}{2} = m$

$m = \frac{10}{2}$

$m = 5g$




5)

Force n pully $= \sqrt{(mg)^2 + (M+m)^2 g^2}$

$= g\sqrt{M^2 + (M+m)^2}$

6)

Let the block moves upward with velocity u,
 ucosθ = v.
 $u = \frac{v}{\cos\theta}$

7)

Let spring force be kx then 210 – kx = 30 $a_{30}$ and kx = 15$a_{15}$ = 15 (6) = 90

So, 210 – 90 = 30$a_{30}$ ⇒ a30 = 4 m/$s^2$

8) **(1) Asking About**
We need to analyze the following:

A. **Assertion:** Whether the apparent weight of a person in a lift moving upwards with uniform acceleration is always higher than their true weight.
B. **Reason:** Whether the weight (gravitational force) always acts downwards is related to the assertion.

**(2) Concept**
**True Weight:** The actual weight of a person is the gravitational force acting on their mass, given by W = mg where m is the mass and g is the acceleration due to gravity.
**Apparent Weight:** The apparent weight of a person in a lift is the normal force (N) exerted by the floor on the person. It changes based on the lift's motion:

A. **Lift at rest or moving at constant velocity:** N - W = mg
B. **Lift accelerating upwards:** N = m(g + a) where a is the upward acceleration.
C. **Lift accelerating downwards:** N = m(g - a)

**(3) Formula**
The apparent weight when the lift moves upwards with uniform acceleration is:
N = m(g + a)
Here:

A. N = Apparent weight (normal force)
B. m = Mass of the person
C. g = Acceleration due to gravity (9.8 m/$s^2$)
D. a = Acceleration of the lift upwards

**(4) Solution / Explanation**
**1. Assertion Analysis:**
When the lift moves upwards with uniform acceleration, the apparent weight is given by N = m(g + a). Since a > 0, the apparent weight is always higher than the true weight W = mg.

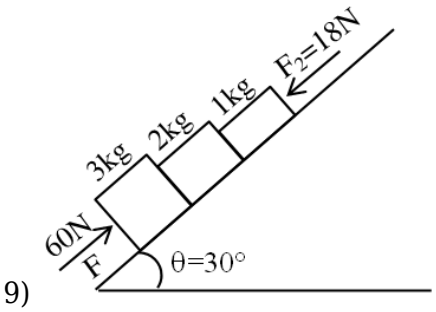
A. **Assertion is True.**

**2. Reason Analysis:**
The reason states that the weight (on Earth's surface) always acts downwards. While this is true, it does not explain why the apparent weight increases when the lift accelerates upwards. The increase in apparent weight is due to the additional force required to accelerate the person upwards.

A. Reason is True but not the correct explanation of the Assertion.

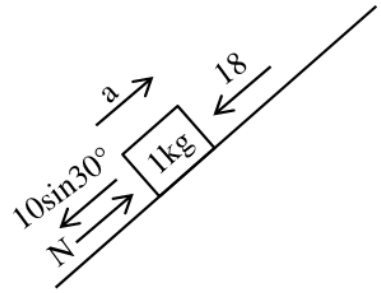
**(5) Final Answer**
**Option 2: Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.**

9)




$$a = \frac{\text{Net force}}{\text{Total mass}} = \frac{60-(18+60\sin 30^\circ)}{6} = 2ms^{-2}$$

FBD of 1kg

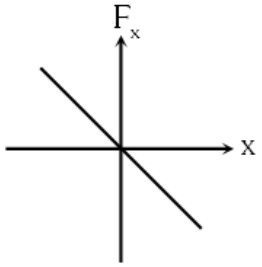



N – 18 – 5 = 1(2)
N = 25 N

10)

Restoring force of spring is
F = – kx

$\boxed{F \propto -x}$




11) $a_{system} = \frac{F}{2M}$ so $T = \left(M + \frac{3}{4}M\right)a$

$T = \frac{7}{4}M\left(\frac{F}{2M}\right) \Rightarrow T = \frac{7F}{8}$

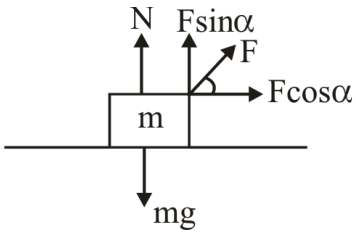
Hence option (1)

12) **Concept :**
Newton's Second Law is only valid in inertial frames (non-accelerating reference frames)

**Formula :** $\vec{F} = m\vec{a}$ This is valid only in an **inertial reference frame**.

**Calculation :**
No numerical calculation is needed : it's theoretical concept.

**Correct Answer : Option (2)** Any Inertial Observer




13)
N + F sin a = mg
when mass leave contact → N $\overline{\text{---}}$ O
F sin α = mg

$$t = \frac{mg}{K \sin \alpha}.$$

14) $T_1 = (12 + 3)a$
$T_1 = 15a$ ...(i)
$T_2 = 3a$ ...(ii)

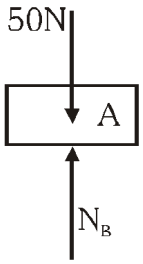
$$\frac{T_1}{T_2} = \frac{5}{1}$$

$T_1 : T_2 = 5 : 1$

15) $$a = \frac{\sqrt{2gh_1} + \sqrt{2gh_2}}{t}$$

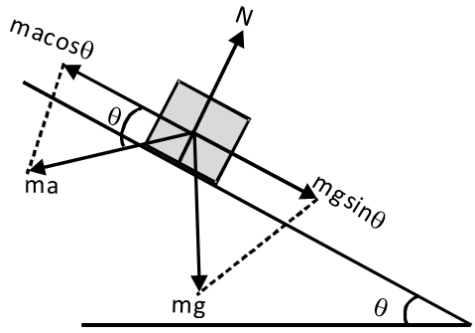
$$a = \frac{\sqrt{2 \times 10 \times 20} + \sqrt{2 \times 10 \times 5}}{0.01}$$

$$= \frac{20 + 10}{0.01} = 3{,}000 \text{ ms}^{-2}$$




16) FBD of block

17) FBD w.r.t. wedge




macosθ = mgsinθ ⇒ a = g tan θ

18) F Δt = change in momentum
|200(0.25)| = |0 – mv|
⇒ mv = 50 kg-ms$^{-1}$
Hence option (1)

19) $a = \frac{600-400}{40} = 5\ m/s^2$
Hence option (1)

20) For 1 kg not slipping
a = g tan θ (θ = 37°)
a = 10 × 3/4
a = 7.5 m/s$^2$
For wedge & 1kg block
T = (1 + 4)a
T = 5 × 7.5
T = 37.5 N
for m
mg – T = ma
10 m – 37.5 = m × 7.5
2.5 m = 37.5
m = 15 kg

21) $T = \frac{2m_1m_2(g+g)}{m_1+m_2} = \frac{4m_1m_2g}{m_1+m_2}$
In terms of w
$T = \frac{4w_1w_2}{w_1+w_2}$

22) Forces from ground causes horse motion.

23) 3g – $T_2$ = 3 a ...(i)
$T_2 - T_1 = 6a$ ...(ii)
$T_1$ – g = a ...(iii)
 By solving eq (i), (ii) and (iii)
2g = 10a

$a = 2\ m/s^2$

24) $a = \frac{60}{12} = 5\ m/s^2$

$T_C = 4 \times 5 = 20\ N$

$T_A = 60 - 6 \times 5 = 30\ N$

$T_B = 5 \times 5 = 25\ N$

25) $\Delta p = 2\ mv\ \cos\theta$

$$F = \frac{dp}{dt} = (2v\cos\theta)\frac{dm}{dt}$$



The mass crossing an area in t would be

$v \times t \times a \times d = m$ ...(1)

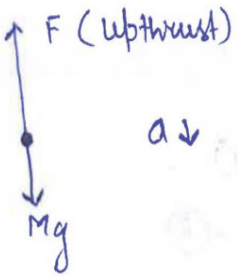
For small element

$\frac{dm}{dt} = vad$

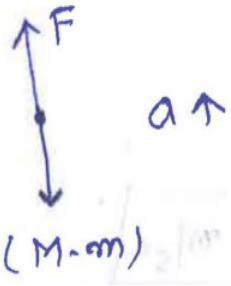
$F = 2av^2d\cos\theta$

26)

A. **Q. Explanation :-** To find out the value of unknown mass
B. **Solution :-**




$Mg - F = ma$ .....(1)




$F - (m - M)g = (m - M)a$ .....(2)

By Eq$^n$ (1) and Eq$^n$ (2)

$$F = mg - ma = (m - M)(g + a)$$

$$\Rightarrow mg - ma - m(g + a) = -M(g + a)$$

$$\Rightarrow M = \frac{2ma}{g + a}$$

A. **Concept :-** Motion of connected bodies.
B. **formula :-** Net force = mass x acceleration

27) $T_2 \sin\theta = T_1 \sin\theta$

$\Rightarrow T_2 = T_1$ & $T_2 = T_3$

28)

Reading of spring balance = 2 × (Tension in string)

In (A) : Reading = $\frac{2(2m_1m_2)}{m_1+m_2}g = 6g$

In (B) : Reading = $\frac{2(2m_1\,m_2)}{m_1+m_2}g = 8g$

29) Inertia of body depends on mass. not on velocity.
Hence option (3)

30) Tension at pt. P = $\left[m + \frac{2L/3}{L}(3m)\right]$ geff.

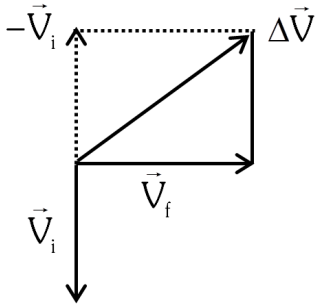
= 3mgeff. = 3m $\left(g + \frac{g}{3}\right)$ = 4mg

31) $\vec{V}_i$ = (V) southward

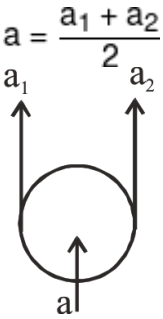
$\vec{V}_F$ = (V) Eastward

$\overrightarrow{\Delta V} = \vec{V}_F - \vec{V}_i$

**= Along North - East**

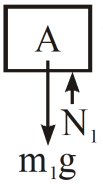



32) For moving pulley

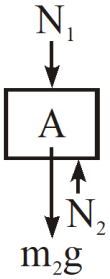



Now according to question

accelertion of block = $\frac{2+0}{2} = 1 m/s^2$




33) $N_1 = m_1g$

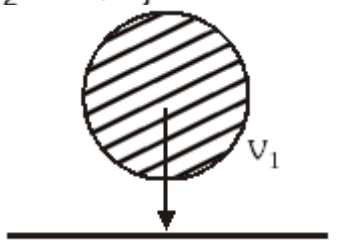



$N_2 = N_1 + m_2g$
$N_1 = (m_1 + m_2)g$
Hence option (2)

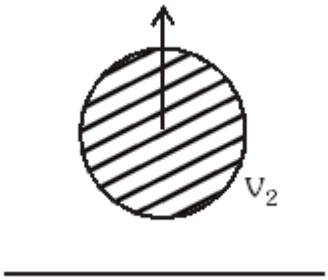
34) $\Delta p = \int F dt = 0$ (area of F-t graph is zero)

35) $F = v\frac{dm}{dt} = 20 \times \frac{50}{60} = 16.66\,N$
Hence option (4)

36) Velocity just before striking the ground
$v_1 = \sqrt{2gh}$
$v_1 = \sqrt{2(10)\,(10)} = 10\sqrt{2}$ m/s
$v_1 = -10\sqrt{2}\,\hat{j}$
If it reaches the same height, speed remains same after collision only the direction changes.
$v_2 = 10\sqrt{2}$ m/s
$\vec{v_2} = 10\sqrt{2}\,\hat{j}$




$|\text{Impulse}| = m|\Delta\vec{v}|$
$= m|10\sqrt{2}\,j - (-10\sqrt{2}\,j)|$
$= 0.15\left[2(10\sqrt{2})\right]$




$= 3\sqrt{2}$ kg m/s
= 4.2 kg m/s

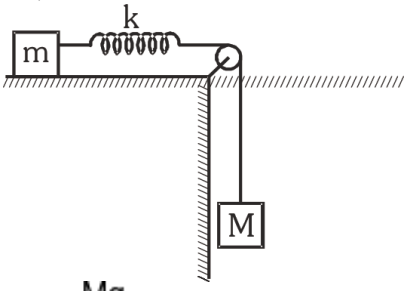
37) T – W = ma
⇒ 5Mg – Mg = Ma
⇒ a = 4g

Hence option (1)

38) $\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$
$\Rightarrow \vec{a} = 0$
$\Rightarrow \vec{v}$ = constant

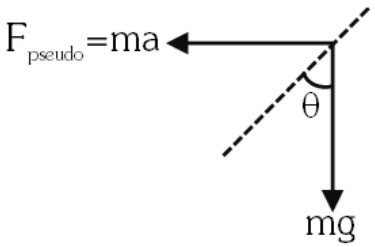
39) T – mg = ma
150 – 100 =10 × a
150 = 100 + 10a
50 = 10a
5 = a
$\frac{g}{2} = a_{min}$

40) Acceleration of blocks




$a = \frac{Mg}{m+M}$
Tension in the string/spring
$T = ma = \frac{mMg}{m+M}$
 Extension in the string, $x = \frac{T}{k} = \frac{mMg}{k(m+M)}$

41) $a = \frac{F}{m}$
$v^2 - 0 = 2\frac{F}{m} \times d$
 $v \propto \frac{1}{\sqrt{m}}$




42)

$$\tan\theta = \frac{F_{pseudo}}{mg} = \frac{a}{g}$$ towards left

43) $F = m_2a_2 = m_1a_1$

$$a_1 = \frac{m_2}{m_1}a_2$$

44)
$$\vec{C} = \vec{A} \times \vec{B} = \begin{vmatrix} i & j & \hat{k} \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = -2\hat{i} = 0\hat{j} + 2\hat{k}$$

$$\vec{C} = \frac{\vec{C}}{|\vec{C}|} = \frac{-2\ell + 2\hat{k}}{\sqrt{4+4}} = \frac{-\hat{i} + \hat{k}}{\sqrt{2}}$$

45)

Option (1) is correct

CHEMISTRY

46)

If proton, neutron, and alpha particles have the same De-Broglie wavelength, the alpha particle will have the highest kinetic energy. The order of kinetic energy from highest to lowest is: alpha particle > proton > neutron
Option (4) is correct

47)

Option (2) is correct

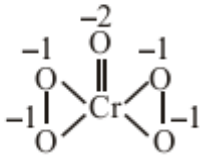
48)
$$As_2^{+6} \longrightarrow 2As^{+5} + 4e^-$$

$$S_3^{-6} \longrightarrow 3s^{+6} \quad \frac{+24e^-}{n-factor = 28e^-}$$

$$E = M/28$$

49) Oxidation number of S are +6, +4, –2, +6 respectively.

The structure of $CrO_5$ is

50) 


51) $\mathbf{n} = |1 + 5 \times 2|$

52)

$[Mo_2O_4(C_2H_4)\ (H_2O)]^{-2}$
$2x + 4(-2) = -2$
$x = +3$

53)

Option (3) is correct

54)

Option (3) is correct

55) $\overset{0}{\underset{6}{C}}H_{12}O_6 \rightarrow \overset{2/3}{\underset{3}{2C}}H_4O_3 + 4H^+ + 4e^-$
n-factor = 4

56)

Option (2) is correct

57)

Option (2) is correct

Mass of $H_2O_2$ = 1gm $\times \left(\frac{x}{100}\right)$ = 0.01

Eq. of $KMnO_4$ = Eq. of $H_2O_2$

58) $N \times V_{(lit)} = \frac{W}{E}$

$N \times \frac{'x'\ ml}{1000} = \frac{(0.01x)gm}{17}$ $(E_{H_2O_2} = \frac{34}{2} = 17)$

$N = \frac{10}{17} = 0.58$

59)

Option (3) is correct

60) $2MnO_4^- + 5H_2O_2 + 6H^+ \longrightarrow 2Mn^{+2} + 8H_2O + 5O_2$

5 mol $H_2O_2$ 5 mol $O_2$

1 mol $H_2O_2$ 1 mol $O_2$

$\frac{1.2 \times 10^{-3}}{34}$ $\frac{1.2 \times 10^{-3}}{34}$ mol of $O_2$

$\frac{1.2 \times 10^{-3}}{34} \times 32$ wt of $O_2$

$1.129 \times 10^{-3}$ gm of $O_2$

61)

Option (3) is correct

62)

Option (4) is correct

63)

Option (2) is correct

64)

Option (2) is correct

65)

**Explaining :** -

**Concept :** This question is based on

**Solution :** 104 unnilquadium
**Final Answer :** (3)

66)

Option (4) is correct

67)

Option (3) is correct

68)

Option (4) is correct

69) Bond length of single covalent bond
$= r_A + r_B - 0.09$ (DEN)
Bond length ($d_{H-Cl}$) = rH + $r_{Cl}$ – 0.09 (DEN)
$r_H$ = 0.37Å ; $r_{Cl}$ = 1.67Å and DEN = 3.0 – 2.1 = 0.9
$d_{H-Cl}$ = 0.37 + 1.67 – 0.09(0.9)
or, $d_{H-Cl}$ = 2.04 – 0.08 = 1.96 Å

70) The atomic radius decreases along the period. Also cations are always smaller than their parent atom and anions are always larger than their parent atom.

71) Lanthanide contraction is observed in these ions, i.e., ionic radius decreases as atomic number increases.

72) In $CrCl_2$, Oxidation state of chromium is +2 (minimum). Thus it will have maximum radius. As positive oxidation state increases, radius decreases.

73)

Option (1) is correct

74)

Option (1) is correct

75) Filling 5d-subshell starts from $6^{th}$ period.

76)

Option (3) is correct

77)

Option (3) is correct

78) Relative positions of these ions in periodic table are :

| | | |
|---|---|---|
| $Li^+$ | $Be^{+2}$ | |
| $Na^+$ | $Mg^{+2}$ | $Al^{+3}$ |

Correct order of ionic radius is
$Na^+ > Li^+ > Mg^{+2} > Al^{+3} > Be^{+2}$
(0.76Å) (0.72 Å)

79)

Option (4) is correct

80) Ta belongs to period VI and group 5 (VB).

81)

Correct $e^-$ configuration is :
$[Kr]5s^2 4d^{10}, 5p^6 6s^2, 4f^{14}$

This element belongs to f-block.

82)

Inert gas elements →$1s^2$ and $ns^2 np^6$
Representative elements → $ns^{1-2}$ and $ns^2np^1$ to $ns^2 np^5$
Transition elements → $(n-1)d^{1-10} ns^{1or2}$

Inner – transition elements → $(n-2)f^{1-14} (n-1)d^{0-1} ns^2$

83)

Bond length of H–F = $r_H + r_F - 0.09 (\Delta EN)$
$= 0.37 + 0.72 - 0.09 (1.9)$

$= 0.92$ Å

84)

The correct order of radius is :

$F < O < F^- < O^{-2}$

85)

Option (2) is correct

86) For 2s, 3p, 4d show same radial probability distribution curve.

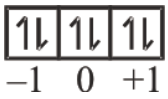
87)

Option (3) is correct

88)

(a)

| | | | | | |
|---|---|---|---|---|---|
| Mg = | $1s^2$ | $2s^2$ | $2p^6$ | $3s^2$ | |
| $\ell$ = | 0 | 0 | 1 | 0 | |
| m = | 0 | 0 | 1 | 0 | for m = 0 |
| | | | | | No. of $e^-$ = 8 |




(b) One orbital occupied maximum $2e^-$

(c) M shell n = 3, no. of $e^-$ = $2n^2 = 2 \times 3^2 = 18$

(d) P = $1s^2 2s^2 2p^6 3s^2 3p^3$

l = 0 for s orbital, so $6e^-$, in s, so for 3e, s = $+\frac{1}{2}$

and for 3e, s = $-\frac{1}{2}$

89)

Option (3) is correct

90)

Option (4) is correct

BIOLOGY

91) NCERT-XI, Pg. # 11

92)

**Explain Question :** Question asking about growth and reproduction in Unicellular organisms.

**Concept :** This question is based on living world

**Solution :** Unicellular organisms shows growth and reproduction through by Cell division

**Final Answer :** option (1)

93)

NCERT (XI) Pg. # 07

94) NCERT (XIth) Pg. # 10

95)

NEW NCERT XI pg no. 6

96) NCERT XI, Pg # 9,10,11

97) NCERT Pg. # 4

98) NCERT XI, Pg. # 5

99)

NCERT XI, Pg. # 9

100) The assertion is true as scientific names are indeed unique and universally accepted to avoid confusion in the scientific community. The reason is also true and is the correct explanation for the assertion because the ICBN and ICZN are the standardized codes that provide the agreed-upon principles and criteria for the naming of plants and animals, ensuring that each organism has a unique name.

101)

NCERT XI Pg. # 9 (Para 3, line 1,2,3)

102)

NCERT XI Pg # 4, 5

103) NCERT-XI, Page- 4,5, Para- 1.1

104) NCERT (XI) Pg. # 5

105) The assertion is true as taxonomy indeed involves characterisation, identification, classification, and nomenclature. The reason is also true and is the correct explanation for the assertion because these taxonomic processes rely on detailed knowledge of the organism's structure, development, and ecology.

106) **Explanation:** The **general characteristics of life** include:
**1. Growth:** Living organisms grow in size or increase in the number of cells.
**2. Reproduction:** Organisms reproduce to create offspring, ensuring the continuity of the species.

**3. Response to stimuli:** Living organisms respond to changes in their environment, such as light, temperature, or touch.
**4. Metabolism:** All living organisms undergo metabolic processes, which involve chemical reactions that provide energy and maintain life.
**5. Cellular organization:** All living organisms are made up of cells, which are the basic structural and functional units of life.

The correct answer is **4. All of these**

107) **The correct answer is A and D.**
Here's why:

A. **A.** Definition of biological species was given by Ernst Mayr: This statement is correct. Ernst Mayr is well-known for his work on the concept of biological species, particularly for defining the species as a group of organisms that can interbreed and produce fertile offspring.
B. **B.** Photoperiod does not affect reproduction in plants: This statement is incorrect. Photoperiod (the length of day and night) does affect the reproduction of many plants, especially in terms of flowering time. Plants are often categorized as short-day or long-day plants, meaning their flowering is triggered by specific light conditions.
C. **C.** Binomial nomenclature system was given by R.H. Whittaker: This statement is incorrect. The binomial nomenclature system (two-part naming system for species) was developed by Carl Linnaeus, not R.H. Whittaker. Whittaker is known for his work on the five-kingdom classification system.
D. **D.** In unicellular organisms, reproduction is synonymous with growth: This statement is correct. In unicellular organisms like amoeba, growth and reproduction are closely linked. When the organism grows, it eventually divides to form offspring through processes like binary fission.

So, the two correct statements are A and D.

108) **Explanation:**
Increase in mass is not exclusive. Non-living things (like crystals) can also increase in mass by surface deposition.
**Correct Answer: 3 – I, III and IV**

109) **Solution:**
**The Correct answer is 2.**
A-4, B-2, C-3, D-5, E-1

110)

NCERT Page No. 6

111)

The correct answer is 1. A-is incorrect and B-is correct.
**Explanation:**

A. Statement A is incorrect because in biological nomenclature, the first word (genus) should

always start with a capital letter, and the second word (species) should start with a lowercase letter. For example, Homo sapiens.

B. Statement B is correct. Biological names are indeed printed in italics (or underlined when handwritten) to distinguish them from other text.

112) In plants, growth by cell division occurs continuously throughout their life span. This continuos growth in plant is axial (i.e., takes place on two axes) and unique. Plant growth consists of primary and secondary growth, on the basis of time when it occurs.

113) Breaking down of sugar into water and $CO_2$ is a catabolic process.

114) Properties of tissues are not present in the constituent cells but arise as a result of interactions among the constituents cells.

115)

The correct answer is **3. Non-livings only.**
**Explanation:** Non-living things such as crystals can grow in size but they **cannot reproduce.** Growth in non-living things occurs due to external factors like temperature or environmental conditions, but they do not have the biological processes required for reproduction, which is a characteristic exclusive to living organisms. Living organisms, on the other hand, can both grow and reproduce. Therefore, non-living things are the correct option that can grow but cannot reproduce.

116) NCERT 11th Class (Chapter-1) Page No. 1

117)

The correct answer is **4. Growth and reproduction.**
**Explanation:** In unicellular organisms, growth and reproduction are mutually inclusive because, in many cases, the process of cell division (reproduction) is synonymous to the increase in cell number (growth). As the organism grows, it prepares to reproduce, often through processes like binary fission.

118) **Correct Answer (4) :** (i), (ii), (iii), (iv) for A(Bacteria) and (iii), (iv) for B(Higher plants).
**Explanation:** Features:

A. **(i) Grow and reproduction both are synonyms:**
    A. **Bacteria (A):** True, as growth and reproduction occur simultaneously in binary fission.
    B. **Higher Plants (B):** False, they grow and reproduce separately.
B. **(ii) Binary fission:**
    A. **Bacteria (A):** True, they reproduce by binary fission.
    B. **Higher Plants (B):** False, they do not use binary fission.
C. **(iii) Cellular organization:**
    A. **Bacteria (A):** True, they are unicellular organisms.
    B. **Higher Plants (B):** True, they have a complex multicellular organization.
D. **(iv) Ability to sense their environment:**

A. **Bacteria (A):** True, they can sense environmental changes.
B. **Higher Plants (B):** True, they also respond to their environment.

119) Isolated metabolic reactions in vitro are not living things, but they are still considered living reactions. All living organisms can sense and respond to their surroundings. Photoperiod affects seasonal breeders, not non-seasonal ones.

120)

NCERT Reference:
Botany-XI Page No.- 5

121)

NCERT XI Pg.# 10,11

122)

NCERT Pg. # 3

123) NCERT Pg. # 8

124) NCERT Pg # 09

125) NCERT-XI, Pg. # 7

126)

The correct answer is 3. **Self organising**, **self regulating interactive system capable of responding to external stimuli.**
**Explanation:**

A. Living organisms are characterized by being **self-organizing** (able to maintain structure),**self-regulating** (able to control internal conditions), and are **interactive systems** (interact with their environment).
B. They also have the ability to **respond to external stimuli** (such as light, heat, or sound), a key feature of life.
C. This matches the definition of living organisms according to basic biological principles.

127) NCERT XI Page No. # 4

128) NCERT XI, Pg. No. 4

129)

**Solution/Explanation/Calculation:**

Metabolism is a characteristic of life, but isolated reactions in a lab do not make a system alive as they lack cellular organization and self-regulation.

**Conclusion:**

Hence, option (4) is correct.  Isolated metabolic reactions *invitro* are living beings.

130)

The correct answer is: **1. Phylum - Class - Order - Family Genus**

**Explanation:** The correct sequence of taxonomic categories in descending order (from the broadest to the most specific) is:
Kingdom > Phylum > Class > Order > Family > Genus > Species
So, the correct sequence starts from Phylum, then goes to Class, Order, Family, and Genus.

131)

NCERT Pg. # 8

132)

**Solution/Explanation/Calculation:**

For bacteria, growth involves both:

A. Increase in mass (the cell accumulates more protoplasm),

B. Increase in number (through binary fission, which is the process of cell division).

C. Regeneration is also method of asexual reproduction in some multicellular organism.

So, the correct answer should be Option 1: iii, iv, v because:

A. iii. Consciousness is the ability of an organism to sense and respond to stimuli. Bacteria do not have consciousness, but they do show a basic response to stimuli (like moving toward nutrients or away from harmful substances).

B. iv. Metabolism is a defining characteristic of life, and bacteria do perform metabolic processes.

C. v. Binary fission is how bacteria reproduce and is a key feature of their growth

**Conclusion:**

Hence, option (1) is correct.

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133) NCERT 11th page # 240 - 241

134)

The Correct answer is: 1
Only (1) is correct.

135)

The correct answer is:
A-Incorrect; B - Correct
**Explanation:**

A. Statement A is incorrect because the properties of tissues do not solely arise from individual cells but are the result of interactions among the cells within the tissue. The structure and function of a tissue depend on how the cells work together.
B. Statement B is correct because the properties of cellular organelles, such as the mitochondria or the endoplasmic reticulum, arise from the interactions among the molecules that make up the organelle, rather than being inherent in the individual molecular component.

136)

NCERT XI Pg#186

137)

NCERT XI Pg#183

138)

NCERT XI Pg#187

139)

NCERT XI Pg#187

140)

NCERT Page No. # 275

141)

NCERT XI Pg#184

142)

NCERT XI Pg#187

143)

**A. Question:** Where is the glottis located?
**B. Given Data:** Options are mouth, trachea, pharynx, and diaphragm.
**C. Concept:** Anatomy of the respiratory system.
**D. Explanation:** The glottis is the opening into the larynx. The larynx is located in the pharynx. Specifically, the glottis is the space between the vocal cords within the larynx. The pharynx is a common passageway for both food and air. The glottis, when open, leads to the trachea and eventually the lungs.
**E. Final Answer:** The glottis is an opening in the floor of the pharynx (specifically, it's the opening to the larynx which is part of the pharynx). Therefore, the correct option is 3.

144)

NCERT XI Pg#184

145)

NCERT XI Pg#187

146)

NCERT XI Pg#184

147)

NCERT XI Pg#184

148)

QUESTION  IS ASKING  TO identify A, B, C, and D when you have the diagram.
provided the image, we can definitively label A, B, C, and D.
Based on the diagram you've given:
A: This represents the enzyme carbonic anhydrase. It facilitates the reaction between $CO_2$ and water to form carbonic acid ($H_2CO_3$).

B: This also represents the enzyme carbonic anhydrase. The same enzyme catalyzes the reverse reaction, converting carbonic acid ($H_2CO_3$) back into $CO_2$ and water.
C: This is $HCO_3^-$, the bicarbonate ion. It's the primary form in which carbon dioxide is transported in the blood plasma.
D: This depicts the chloride shift or Hamburger phenomenon. As bicarbonate ions ($HCO_3^-$) leave the red blood cell, chloride ions ($Cl^-$) from the plasma enter to maintain electrical neutrality.

Therefore, the correct option is:

2. A - Carbonic anhydrase, B - Carbonic anhydrase, C - $HCO_3^-$ and D - Hamburger phenomenon

149)

NCERT XI Pg#190

150)

NCERT XI Pg#186

151) NCERT XI, Page # 273

152) NCERT XI, Fig. 17.5

153)

NCERT XI Pg#189

154)

NCERT XI Pg#188

155)

NCERT XI Pg#190

156) NCERT(XI) 17.5

157)

NCERT XI Page # 270

158)

(a) High $H^{\oplus}$ concentration and higher temperature are favourable for dissociation of oxygen

from oxyhaemoglobin. Correct. This describes the Bohr effect. Increased H+ concentration (lower pH) and higher temperature both decrease the affinity of hemoglobin for oxygen, causing it to release oxygen more readily. This is important in metabolically active tissues where these conditions prevail.
(b) Partial pressure of oxygen in alveoli is higher than in Tissue. Correct. The alveoli are where oxygen is loaded into the blood. For oxygen to diffuse from the alveoli into the blood and then to the tissues, the partial pressure of oxygen ($PO_2$) in the alveoli must be higher than in the tissues. This pressure gradient drives oxygen transport.
(c) The maximum volume of air a person can breathe in after a forced expiration is vital capacity. Correct. Vital capacity (VC) is the total volume of air a person can exhale after a maximal inhalation. It represents the total exchangeable air in the lungs.
(d) In alveoli $PCO_2$ low and $PO_2$ high are for dissociation of $CO_2$ from carbamino-haemoglobin. Correct. In the alveoli, the partial pressure of carbon dioxide ($PCO_2$) is low, and the partial pressure of oxygen ($PO_2$) is high. These conditions favor the release of $CO_2$ from carbaminohemoglobin (hemoglobin bound to $CO_2$) and the binding of oxygen to hemoglobin.

159)

NCERT XI Pg#189

160)

A. Question: Assertion: Larynx is a cartilaginous box which helps in sound production. Reason: Epiglottis flap covers the glottis during swallowing and prevents the entry of food into the larynx.
B. Given Data: Assertion and Reason statements about the larynx and epiglottis.
C. Concept: Anatomy and function of the larynx and epiglottis.
D. Explanation:

A. Assertion: The assertion is true. The larynx is indeed a cartilaginous structure, and it houses the vocal cords, which vibrate to produce sound.
B. Reason: The reason is also true. The epiglottis is a flap-like structure that covers the glottis (the opening of the larynx) during swallowing. This action prevents food from entering the trachea and larynx.
C. Relationship: While both statements are true, the reason is not the direct explanation for the assertion. The larynx produces sound because of the vocal cords, not because of the epiglottis's role in swallowing. The epiglottis's function is to protect the airway, not to produce sound.

E. Final Answer: 1. Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

161)

NCERT XI Pg#189

162)

NCERT XI Pg#185

163) NCERT (XI) Pg. # 274

164)

NCERT XI Pg#190

165) NCERT (XI$^{th}$) Pg. # 275, Para-4

166)

NCERT XI Pg#187

167)

NCERT XI Pg#189

168) NCERT (XI$^{th}$) Pg. # 275

169) NCERT-XI, Pg#276

170)

NCERT XI Pg#186

171)

NCERT XI Pg#187

172) NCERT page: 272

173) NCERT XI Page 190

174) NCERT XI Page -186,187

175)

NCERT XI Pg#189

176)

NCERT XI Pg#187

177)

NCERT XI Pg#183

178)

NCERT XI Pg#190

179)

NCERT XI Pg#187

180)

NCERT XI Pg#183